MILITARY SPECIFICATION

INTRAVENOUS INJECTION SET, Secondary Non-Vented With Flexible Drip Chamber, Disposable

This specification is approved for use within the Defense Personnel Support Center, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a sterile, disposable, non-vented secondary intravenous injection set (infusion apparatus) designed for use in the administration of parenteral solutions from standard glass solution containers without airways.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 S. 20th Street, Philadelphia, PA 19101, by using self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6515
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SPECIFICATIONS

FEDERAL

GG-N-196	- Needles, Hypodermic, for Luer-Syringe.
MMM-A-250	- Adhesive, Water-Resistant, (for Sealing Fiberboard Boxes).
PPP-B-601	- Boxes, Wood, Cleated-plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-676	- Boxes, Set-up.
PPP-T-76	- Tape, Pressure-Sensitive Adhesive Paper, (for Carton Sealing).

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-L-10547	- Liners, Case and Sheet, Overwrap; Waterproof for Vaporproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-147	- Palletized Unit Loads.
MIL-STD-794	- Part and Equipment, Procedures for Packaging and Packing of.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

U.S DEPARTMENT OF HEALTH, EDUCATION AND WELFARE, Bureau of Biological Food and Drug Administration Code of Federal Regulation Part 273. Minimum Requirements (as applicable).

(Application for copies should be addressed to the Food and Drug Administration, 5600 Fishers Lane, Rockville, MD 20852).

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/HIMA MD70.1 - LUER TAPER FITTINGS

(Application for copies should be addressed to the Publication Sales Dept., STD, American Institute of Physics, 335 E. 45th Street, NY, NY 10017).

THE UNITED STATES PHARMACOPEIAL CONVENTION, INC.
The Pharmacopeia of the United States of America (USP).

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM) STANDARDS

F-899 - Stainless Steel Billet, Bar and Wire for Surgical Instruments

(Application for copies should be addressed to he American Soceity for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 Material.

3.1.1 Rubber and plastic components. All rubber and plastic components shall be of the best quality and suitable in all respects for the intended purpose. All rubber and plastic components shall be sterilizable, free from pyrogens, compatible, and safe when tested as specified in section 4. Those rubber and plastic components coming in contact with solutions being administered, and end protectors, shall contain not more than 5 parts per million (p.p.m.) total of lead, barium and cadmium when tested as specified in 4.5.1, or 4.5.1.1. The plastics shall meet the requirements of the Pharmacopeia of the United States (USP) for Class IV plastics and for transfusion and infusion assemblies when tested as specified in 4.5.2.

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3.1.1.1 Plastic tubing. Plastic tubing shall be clear, colorless, transparent, flexible polyvinyl chloride, having a hardness of 68 + 6, -5 Shore A durometer, when tested as specified in 4.5.3. The tubing shall be flexible and pliable, but sufficiently rigid to prevent collapse during normal use. The tubing shall be clean and free of particles and shall be free from all liquid substances which oxidize, exude, or sweat-out to the surface of the tubing.

3.1.2 Needles.

3.1.2.1 Cannulae of needles. Cannulae of needles shall be fabricated from type 304 stainless steel in accordance with ASTM F899, and shall meet the corrosion-resistance, elasticity, and resistance-to-breakage tests as specified in GG-N-196. The cannulae shall be fabricated from either seamless drawn, or welded and drawn tubing.

3.1.2.2 Hub. Hubs shall be fabricated from a suitable plastic (containing no exudable toxic materials), aluminum alloy, nickel or chromium-nickel plated brass. Aluminum hubs shall be degreased and then either anodized or chemically passivated.

3.1.3 Other material. All other material hereinafter specified shall be of the best quality and suitable in all respects for the purpose intended.

3.2 Injection set. The set shall be a sterile, non-vented, non-pyrogenic, safe (non-toxic) disposable secondary intravenous injection set (infusion device) for use in the administration of parenteral solutions from either standard glass solution container with an airway or a flexible plastic bag. The set shall have a piercing device, a flexible plastic calibrated drip chamber, a length of plastic tubing, a male Luer adapter, an intravenous needle, a roller-type flow control device and end protector. The set shall have an overall length of 36 inches + 6 inches, -1 inch, measured from the tip of the piercing device to the end of the male adapter. The drip chamber shall be molded of flexible, colorless, clear plastic. The piercing device shall be permanently attached to the drip chamber in a leakproof manner. The assembled drip chamber shall be so shaped as to provide a convenient hand grip for the insertion of the piercing device into the stopper or port of the solution container. The assembly set shall be structurally sound and shall not leak air or fluid during use. The open ends of the set shall be covered (sealed) with readily removable covers. A suspension hook hanger approximately 9 inches long designed to lower the primary intravenous set below the secondary set shall be included with the injection set.

3.2.1 Drip chamber. The drip chamber housing shall be an overall length of 2-7/8 inches ± 1/4 inch, measured from the top of the open drip barrel to the bottom of the tubing connector end. The inside diameter at the bottom stem, where the plastic tubing connects, shall be 1/16 to 9/64 inch. The volume capacity of the drip chamber measured up to the level of the drip orifice shall be 3 to 7 ml.

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3.2.2 Cap (piercing device). The cap shall consist of an integrally molded unit including the piercing device (spike). The piercing device shall be manufactured from a suitable rigid, plastic, conical spike. The piercing device tip shall be 7/8 to one inch in length with a diameter of 0.258 to 0.267 inch at the base. The piercing device tip shall be sharpened to such a point as to be suitable for puncture of and inserting into the stopper or port of the standard solution container without undue pressure. Edges of the fluid inlet passages shall be rounded or dulled to prevent stopper coring. The piercing device shall be ruggedly designed to prevent coring or shearing. It shall be firmly retained in a leakproof manner when inserted into the stopper or port of a standard solution container. The piercing device shall not break or leak air or fluid during use. It shall not be brittle or soft, or become brittle or soft when under normal storage conditions. The fluid passage in the spike shall extend into the drip chamber and shall be calibrated to deliver 15 drops per milliliter (ml) when tested as specified in 4.5.4. The device shall maintain 15 drops per ml drop size during the administration of intravenous solutions for a minimum of 48 hours. The drop size of each set shall not vary by more than plus or minus 3 drops per ml.

3.2.2.1 Durability of piercing device. When tested as specified in 4.5.5, the piercing device or cap shall not loosen or permit leakage, nor shall the device become damaged. The piercing device shall be capable of being inserted into a standard solution container without difficulty. The tip or point of the device shall not break or bend to a degree resulting in difficulty in insertion.

3.2.3 Plastic tubing. The plastic tubing shall be fabricated of material as specified in 3.1.1.1. The tubing shall have an inside diameter of 0.100 to 0.110 inch and an outside diameter of 0.140 to 0.155 inch. It shall be firmly and permanently secured, in a leakproof manner, to the drip chamber at one end and to the male needle adapter at the other end. The tubing shall form an airtight, leakproof connection which will not separate when tested as specified in 4.5.6.

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3.2.4 Control clamp. The control clamp shall be fabricated of plastic and shall be a roller-type clamp from which the roller cannot be accidentally removed. The clamp shall be capable of withstanding a 3.65 kilogram load when tested as specified in 4.5.8 without the roller separating from the body of the clamp. The clamp shall be supplied on the tubing in the open position. It shall provide a constant setting and a quick shut-off from the operating position. The clamp shall be movable and shall be capable of one-handed operation. To assure the constant delivery of fluid, it shall be impossible for the clamp setting to be drastically changed by an accidental jerk on the tubing. When tested as specified in 4.5.8, the clamp shall be capable of accomplishing complete quick shutoff (i.e., no falling drip in the drip chamber or from the needle adapter), shall regulate the flow of the solution from complete shut-off to maximum flow and with a constant head of water shall maintain a constant flow rate at any point between complete shut-off and maximum flow. In addition, when the clamp is operated from it fully open position to complete shut off, and this operation is repeated ten times while the clamp is in the same position on the tubing. The tubing shall not be cut, pierced or otherwise damaged so as to cause leakage or restriction of the flow of liquid.

3.2.5 Male needle adapter. The male needle adapter shall be fabricated of plastic and shall be tapered to fit a needle having a standard Luer hub in accordance with ANSI/HIMA Standard MD70.1 in a tight leakproof manner. The needle adapter shall not separate from the tubing or show evidence of leakage when tested as specified in 4.5.6.

3.2.6 Intravenous needle. The needle shall be an 18 or 19 gage, 3/4 to 1 inch cannula intravenous short bevel needle. Except as specified herein, the needle shall be in accordance with the requirements of GG-N-196 for type I, point style b, hub style 1 or 2. The cannula and hub shall be fabricated of material as specified in 3.1.2. The hub cavity shall be constructed in accordance with ANSI/HIMA Standard MD70.1. The needle shall fit the needle adapter in such a manner as to make a tight, leakproof connection. The hub cavity shall be tested for diameter depth and fluid tightness as specified in 4.5.9. Hubs shall fit the plug gage within the "GO" - "NO GO" limits of ANSI/HIMA MD70.1. Hubs which do not fit the plug gage within these limits shall be rejected. When the hub is fitted onto the plug gage as specified in 4.5.9.1, there shall be no movement of the hub when the cannula is pressed lightly from side to side in all directions. When tested for fluid tightness as specified in 4.5.9.2, the swaged connection of the cannula with the hub and the connection of the hub with the test gage shall not show leakage exceeding the formation of a falling drop. The needle shall be supplied either attached or unattached to the tapered end of the needle adapter. Should the needle be supplied unattached, it shall be encapsulated with end protectors, easily removable for attachment to the needle adapter and designed to retain its sterility. The needle marking required by GG-N-196 may be omitted.

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3.2.7 End protectors. End protectors shall be supplied on each open end of the injection set. The end protector shall be attached to the needle adapter in such a manner that will prevent contact of the needle point with the inside of the protector when the needle is attached to the injection set. End protectors shall be readily removable, plastic and constructed in such a manner that when placed over each open end of the injection it shall, after sterilization, maintain sterility of the needle and the entire fluid path of the device.

3.3 Manufacturer's record. The manufacturer shall place a lot control number on the paper label or on the printed carton which can be verified against the manufacturer's records of all tests made available for inspection by any duly qualified medical officer of the Armed Forces, any official of the Food and Drug Administration, or any other duly authorized Government representative. No shipment shall be made until all tests have been completed, verified and recorded in the permanent record of the manufacturer.

3.4 Sterilization, safety and freedom from pyrogens. All equipment shall be safe. Fluid paths, end connectors and needles shall be sterile and pyrogen free. The fluid path, end connectors and needles shall be pyrogen free when tested as specified in 4.5.10 and sterile when tested as specified in 4.5.11. The sets shall be safe when tested for safety as specified in 4.5.12.

3.5 Certificates. Certificate of sterility, freedom from pyrogens and safety shall be furnished to the procuring activity via the government inspector.

3.5.1 Certificate of sterility. A certificate of sterility, listing all sterility lot or control numbers shall state that samples from each sterility lot or control have been tested for sterility as required in 3.4 and were found to be sterile. The certificate shall reference the method used in determining sterility.

3.5.2 Certificate of freedom from pyrogens. A certificate of freedom, listing all lot or control numbers, shall state that samples from each lot or control have been tested for pyrogens as specified in 3.4 and were found to comply with such requirements. The certificate shall reference the method used in determining freedom from pyrogens.

3.5.3 Certificate of safety. A certificate of safety, listing all lot or control numbers, shall state that samples from each lot or control have been tested for safety as specified in 3.4 and were found to comply with such requirements. The certificate shall reference the method used in determining safety.

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3.6 Compatibility. The materials of construction of each intravenous injection set shall not react with, or be acted upon by, the liquids with which they will be in contact during normal use and operation. There shall not be any materials incorporated into the set which will cause any deleterious effect when the set is used for the purpose intended.

3.7 Leakage and separation of set. The set shall be tested for leakage and separation of the set as specified in 4.5.6. Sets which show leakage or separation at any part of the set shall be rejected. A leak is defined as the formation of an air (gas) bubble or liquid drop as appropriate, which breaks away from the set within the specified test time.

3.8 Solution flow rate. When tested as specified in 4.5.7, the intravenous injection set shall permit the passage of 100 cc of fluid in not more than 5 minutes.

3.9 Instructions. Each individual set shall have detailed instructions for the proper use of the set for single use. Detailed instructions for tandem use shall be included in each unit package. The instructions shall include the drop size (number of drops per ml).

3.10 Workmanship. Workmanship shall be first class throughout. The intravenous injection set shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements sections 3 and 5. The inspection set forth in this specification shall become part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Inspection of packaging for delivery. The sample unit shall be one exterior container. The sampling unit shall be in accordance with MIL-STD-105. The inspection level shall be S-4 with AQL of 2.5 expressed in terms of percent defective. The inspection of the packaging, packing and marking for shipment and storage shall conform to the quality assurance provisions of the applicable container specification and marking requirements of MIL-STD-129.

4.2 Sampling.

4.3.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in Table I. The unit of product for examination shall be one intravenous injection set.

TABLE I. Sampling for examination

Type of defect	Inspection level	AQL (percent defective)*
Major	S-4	1.0
Minor	S-4	2.5

4.3.2 For tests. Sampling for tests shall be in accordance with MIL-STD-105 and as indicated in table II. The unit of product for test purpose shall be one intravenous injection set.

TABLE II. Sampling for other tests

Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Heavy Metal ** (extractable)	3.1.1	4.5.1	S-1	1.0
Class IV Plastic **	3.1.1	4.5.2	S-1	1.0
Tubing Hardness	3.1.1.1	4.5.3	S-2	1.0
Drop test	3.2.2	4.5.4	S-3	1.0
Piercing device	3.2.2.1	4.5.5	S-1	1.0
Control Clamp	3.2.4	4.5.8	S-2	1.0
Intravenous needle	3.2.6	4.5.9	S-1	1.0
Freedom from pyrogens	3.4	4.5.10	S-1	*
Sterility	3.4	4.5.11	See 4.5.11	*
Safety	3.4	4.5.10	S-1	*
Leakage and separation of set	3.7	4.5.6	S-3	1.0
Solution flow rate	3.8	4.5.7	S-3	1.0

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* Acceptance number shall be zero.

** Each lot of product need not be tested provided evidence is furnished that samples of the formulation and identical method of processing have previously been tested, that, they comply with USP requirements for Class IV plastics and heavy metals and that they are the same as being offered on this contract.

4.3.3. Sample unit usage. Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.4 Examination. The sets shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted in accordance with MIL-STD-105 and the classification of defects specified in Table III. Examination, however, shall not be restricted to the classified possible defects listed in Table III.

TABLE III. Classification of defects

Categories	Defects
Critical	None defined
Major	
101	End protectors or other components missing or dislodged.
102	End protectors not properly designed to permit flow of sterilizing gases, if used.
103	Not free of bloom or tackiness.
104	End protectors not properly designed to maintain sterility.
105	Style and design not as specified.
106	Presence of foreign matter.
107	Set not free of holes or cuts.
108	Dimensions not as specified.
Minor	
201	Packaging, packing and marking not as specified.
202	Instructions missing, incomplete or illegible.
203	Contractor's name or registered trademark, missing, incomplete, or illegible.

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4.5 Tests.

4.5.1 Heavy metals extractable test. A 2-gram composite sample shall be obtained from the rubber and plastic components. The material shall be cut into small slivers, so that no thickness exceeds 3 mm, and eluted for 72 hours at 50°C, 24 hours at 70°C, 60 minutes at 121°C in one menstruum of 0.9% sodium chloride. Each gram of the material shall be covered with 5 milliliters (ml) of the menstruum. The menstrua are collected, agitated after elution and then submitted to spectographic absorption, spectrophotometric or specific wet analysis for lead, barium, and cadmium. The metals are reported in parts per million. Thus 1.0 ppm in elute is equivalent to extracting 1.0 mg from 200 g of test material dispersed in 1,000 ml of elution menstruum.

4.5.1.1 Alternate heavy metals extractable test. Accurately weigh a 5-gram sample for each of the metals (barium, cadmium or lead) into a 1-oz. Vycor crucible. Reduce to ash in a muffle furnace at 650°C for eight hours. Let cool to room temperature and add 15 ml of 1:1 hydrochloric acid. Bring to a boil on a hot plate, allow to boil to 2 to 3 minutes, remove, let cool, and filter through Whatman #2 filter paper into a 50-ml volumetric flask. Rinse the residue on the filter with deionised water and dilute to 50 ml. Prepare standard solutions of 1, 2 and 5 ppm for each of the metals, namely barium, cadmium, and lead. Determine the absorbances of these solutions at appropriate wave lengths using a Hollow Cathode Ba, Cd, or Pb lamp for each of the respective metals and 10X scale expansion on a spectrophotometer, such as the Perkin-Elmer Model 303, Atomic Absorption Spectrophotometer. Determine the absorbance of the sample solutions using the same conditions as with the standards.

Calculation:

$$\frac{A \times F \times \text{dilution}}{W} = \text{ppm barium, cadium, or lead}$$

A = Absorbance
F = Factor = $\frac{\text{(Concentration standard in ppm Ba, Cd, or PB)*}}{\text{absorbance}}$
W = Sample weight in grams

*The three standards are averaged to give ppm/unit A.

4.5.2 Biological tests. The plastics shall be tested in accordance with the USP for Class IV plastics. Sampling shall be as shown in Table II.

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4.5.3 Tubing hardness. Plastic tubing hardness shall be determined in accordance with ASTM D2240 Standard Test Method for Rubber Property - Durometer Hardness. The test shall be conducted at a room temperature of 27° plus or minus 5°C (80° plus or minus 9°F) on samples which had been conditioned at this temperature for not less than 3 hours prior to testing. The test sample shall be laminated to a thickness of not less than 1/4 inch. The gage reading shall be made 15 seconds after application of the instrument.

4.5.4 Drop test. The assembled set, without the needle attached, shall be connected to a standard solution container filled with either saline solution of water or a 5% Dextrose solution, USP. The container shall be suspended in the normal manner at a sufficient height to permit the entire length of tubing of the set to hang vertically. Establish a flow of 75 $\pm$ 5 drops/minute. After 2 to 5 minutes recheck and, if required, reset the flow rate. After one hour, count the number of drops required to collect 3 mls of sultion.

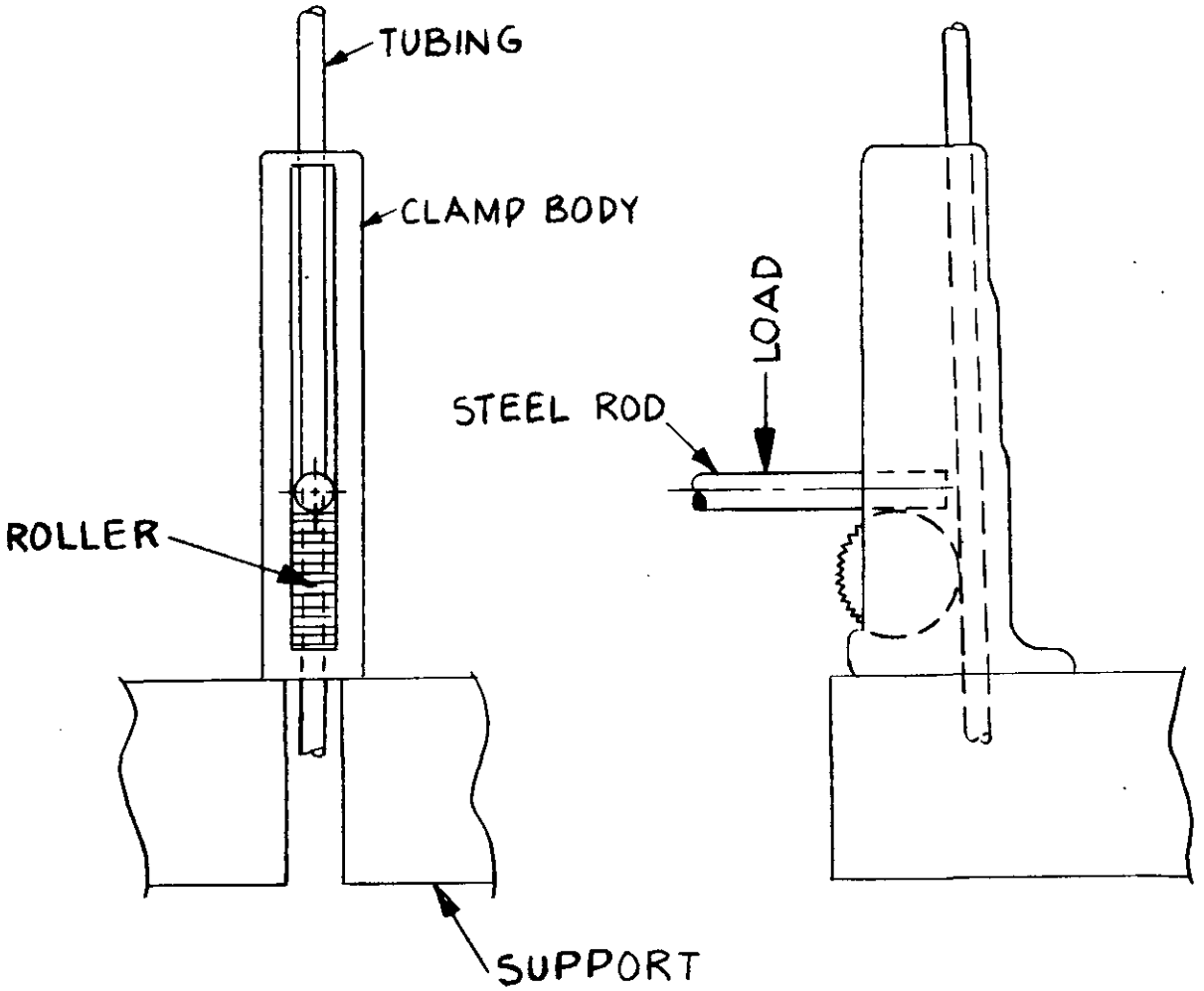
4.5.5 Piercing device durability test. Samples of finished sets, after sterilization, shall be used for this test. The piercing device while attached to the injection set, shall be inserted into and removed from a standard solution container in the normal manner. This operation shall be repeated twice. The piercing device shall then be immersed in a solution of 5 percent dextrose in isotonic sodium chloide solution, in accordance with the requirements of the USP monograph for "Dextrose and Sodium Chloride Injections," for 4 hours. The piercing device shall then be removed from the solution and immediately inserted into a standard solution container.

4.5.6 Leakage and separation of set. The test set shall be applied to those samples subjected to the piercing device durability test (4.5.5). The set shall be immersed in water and subjected to a pneumatic pressure of 8 pounds per square inch gage (psig) for 60 seconds The pressure shall be applied through the piercing device and the set observed for separation and leakage. There shall be no more than one clinging air bubble.

4.5.7 Solution flow rate. The finished intravenous injection set shall be connected to a standard one liter glass solution bottle suspended from a standard IV pole at a height of 36 inches. There shall be no restriction of flow imposed by the clamp or needle. The time necessary for 100 cc of sterile water to pass through the entire set shall be determined.

4.5.8 Clamping device performance test. The assembled set without the needle attached, shall be connected to a standard solution container filled with either a saline solution of water or a 5% Dextrose solution, USP. The container shall be suspended in the normal manner at a sufficient height to permit the entire length of

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METHOD OF APPLYING LOAD TO ROLLER

FIGURE 1

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tubing of the set to hang vertically. The ability of the clamp to accomplish complete shutoff, to regulate flow and with a constant head of liquid maintain a constant rate of flow, shall be tested with the clamp at a point on the plastic tubing within two inches above the male needle adapter. The clamp shall meet the requirements as specified in 3.2.4. The tests shall be performed with the tubing hanging vertically at all times. The rate of flow shall be determined either by counting the drops in the drip chamber during a measured time period or by recording the time required to collect a measured volume of fluid from the needle adapter.

4.5.8.1 Security of roller in clamp body. The roller clamp shall be removed from the set together with a piece of tubing approximately one inch longer than the clamp body. The tubing shall be present in the roller clamp in its normal position with a 1/2 inch length extending from each end of the roller clamp. The clamp, see Figure 1, shall be held vertically, with the large end down, over a supporting gap so that the roller, if forced out of the body, is free to fall. A polished steel rod is applied to the top of the roller, so that the roller is able to rotate in the course of the test and the vertical downward force required to separate the roller from the clamp body is measured. As an alternate, any position of the clamp is acceptable as long as the clamp is not distorted and the load measurement is accurate.

4.5.9 Intravenous body needle.

4.5.9.1 Diameter and depth of hub cavity: The needle hub cavity shall be tested on the male Luer taper gage in accordance with ANSI/HIMA MD70.1. This gage shall be as shown in ANSI/HIMA MD 70.1 with a suitable fluid path to permit performance of the fluid tightness test specified in 4.5.9.2. The fit of the hub on this gage shall be tested in accordance with ASNI/HIMA MD70.1.

4.5.9.2 Fluid tightness. With the needle in horizontal position, the hub shall be attached by means of a plug gage, as shown in Figure 6 of ANSI/HIMA MD70.1, to a water pressure source having a pressure gage. The water shall be run through the needle so as to fill any hub cavity space and the lumen of the cannula. The lumen shall then be sealed by a piece of solid rubber slipped onto the point and a pressure of 45 pounds per square inch applied for 10 seconds.

4.5.10 Freedom from pyrogens. Samples of the finished sets shall be tested for freedom from pyrogens, in accordance with the USP. Sampling shall be shown in Table II.

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4.5.11 Sterility test. The intravenous injection sets shall be tested for sterility in accordance with the method specified in the USP using inoculated product, inoculated simulated product, or paper spore strip (carrier). Product containing a biological indicator shall be marked or labeled so that it clearly differentiates it from product intended for distribution. The incubation medium for the paper spore strip (carrier) shall be soybean-casein digest. The incubation media for the inoculated product, or inoculated simulated product shall be fluid thioglycollate or alternative thioglycollate at 30° to 35°C for one half of the specimens and soybean-casin digest at 20° to 25°C for the remaining half of the specimens. The minimum incubation time shall be ten days. Sampling shall be as follows:

Medium	Paper Spore Strip	Inoc. Prod. or Inoc. Simulated Prod.
Soybean-casein digest	S-3*	S-3*
Fluid thioglycollate or Alternative thioglycollate		S-3*

* Use a minimum of 20 specimens. Acceptance number is zero.

4.5.11.1 Inoculated product or simulated product. For each type of sterilization, the organism used shall be as listed below:

Type of Sterilization	Organism
Steam	B. stearothermophilus
Dry heat	B. stearothermophilus
	B. subtilis
Ethylene oxide	B. subtilis
Gamma radiation	B. pumilus

The number of viable spores shall be sufficient to provide assurance that the product has been sterilized.

4.5.11.2 Inoculated carrier. Inoculated carriers (paper spore strips) if used, shall be in accordance with MIL-S-36586, Sterilization Test Strip Set, Bacterial Spore. Other paper carriers shall be inoculated to be of equivalent resistance. The method of incubation and number of test units shall be as indicated in 4.5.11 Sterility test and 4.5.11.1 Inoculated product or simulated product. The method of sterilization will control the organism in the spore strip to be tested for as shown in 4.5.11.1. NOTE: When an inoculated carrier is chosen, product samples must also be tested as required by the USP and paragraphs 4.5.11 and 4.5.11.1.

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4.5.11.3 Alternate method. Suppliers conducting sterility tests or designating lots in a manner other than that indicated herein shall submit proof that the alternate method provides equal assurance concerning the sterility of the product. However, all sterility tests shall be conducted using biological indicators. Requests for approval of any alternate method of testing for sterility shall be made to the Contracting Officer and approval obtained before employment of the alternate method on a government contract.

4.5.11.4 Product testing. Testing for sterility of the product after receipt by the government shall be without a biological indicator and in accordance with USP. The inspection level shall be as stated in paragraph 4.5.11. Acceptance number shall be zero.

4.5.12 Safety test. The finished sets shall be tested for safety in accordance with the USP. Sampling shall be as shown in Table II.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Individual package. Each set shall have the tubing coiled in such a manner to prevent kinking. Each set shall be further packaged in a sealed paperboard box or in a rigid plastic tray having a tyvek closure. As an alternate, each set shall be packaged in a paperboard tray or box having a die cut window. The tray or box shall be wrapped in clear polyester. The wrap shall be sealed and fitted with a tear strip for ease in opening.

5.1.1.2 Unit package. Fifty individually packaged intravenous injection sets shall be packaged in a box conforming to PPP-B-636, class domestic. Closure shall be effected with tape or adhesive as specified in the box spcification. Metal stitches shall not be used.

5.1.2 Level C. The level C preservation for intravenous injection sets shall conform to MIL-STD-794 requirements for this level.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Four unit packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the applicable box specification.

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5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof liner conforming to MIL-L-10547. Closure and sealing shall be as specified in the liner specification. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in PPP-B-636.

5.2.2 Level B. Four unit packages shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-601, domestic; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the applicable box specification.

5.2.3 Level C. The level C packing for intravenous injection sets shall conform to the MIL-STD-794 requirements for this level.

5.2.4 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in a exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which will insure safe delivery to destination.

5.2.5 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Loads shall be unitized on pallets conforming to MIL-STD-147. Quantities less than 40 cubic feet need not be unitized. Pallet loads including the pallet shall not exceed 43 inches in height, 48 inches in width, and 40 inches in length, and shall conform to Type IV or Type V.

5.2.5.1 Levels A and B. Intravenous injection sets packed as specified in 5.2.1 and 5.2.2 shall be unitized on pallets as specified in 5.2.5.

5.2.5.2 Level C. Intravenous injection sets, packed as specified in 5.2.3, shall be unitized as specified in MIL-STD-794.

5.3 Marking.

5.3.1 Levels A, B and C.

5.3.3.1 Individual package. Each package shall be marked in accordance with commercial practice and the requirements of the Federal Food, Drug, and Cosmetic Act. In addition, each unit shall contain instructions as specified in 3.9 and the legend: "Fluid path and end connects are sterile and pyrogen free." Marking shall include the lot and control number.

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5.3.1.2 Unit package. Each unit package shall be marked as specified in MIL-STD-129. Marking shall indicate the lot number.

5.3.1.3 Exterior container and unitized load. Each exterior container and unitized load shall be marked as specified in MIL-STD-129. Marking shall include the lot number.

5.4 General.

5.4.1 Exterior container. Exterior container (see 5.2.1, 5.2.2, and 5.2.3) shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.2.

6. NOTES

6.1 Intended use. The intravenous injection set is intended for accurate administration of parenteral solutions from standard glass solution containers.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Applicable levels of packaging and packing (see 5.1 and 5.2).
(c) National Stock Number.

6.3 National Stock Number (NSN) coverage. This specification covers the following item:

NSN	ITEM IDENTIFICATION
6515-01-153-5669	INTRAVENOUS INJECTION SET, Secondary, with Flexible Drip Chamber, Non-Vented, 18 Or 19 Gage Needle, Disposable 48-50s

6.4 Subject term (key word) listing.

Intravenous Injection set
Non-Vented
Secondary

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MILITARY INTERESTS:

Custodians:

Army - MD
Navy - MS
Air Force - 03

Review Activities:

Army - MD
Navy - MS
Air Force - 03

Preparing Activity:

DoD-MB

Agent:

DLA-DM

Project No. 6515-3431

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-37990 (DLA-DM)	Intravenous Inejection Set

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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